

Early Detection of Student Mental Health and Academic Burnout Using Multimodal AI-Based Behavioral, Physiological, and Emotional Analysis

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Abstract—Mental health issues such as stress, anxiety, depression, and academic burnout are increasingly common among university students and have a significant impact on academic performance and long-term well-being. Existing assessment approaches rely mainly on self-reported questionnaires and periodic evaluations, which are reactive, subjective, and ineffective for early intervention. This paper presents a multimodal artificial intelligence-based system for early identification of student mental health conditions by analyzing behavioral, physiological, emotional, and academic data. The proposed framework integrates facial emotion recognition, wearable sensor data analysis, natural language processing of reflective text to continuously monitor student well-being in a privacy-aware manner. Machine learning and deep learning models are employed to detect stress, anxiety, and burnout indicators and to predict future mental health risks. Experimental results obtained from real and synthetic datasets demonstrate that multimodal analysis provides more reliable and accurate predictions than single-source methods. The proposed system enables early risk identification and supports timely intervention in academic environments.

Keywords—student mental health, burnout detection, multimodal artificial intelligence, wearable sensors, facial emotion recognition, behavioral analysis (key words)

I. INTRODUCTION

One of the issues of worldwide concern has been in the area of mental health in university students. Modern students are also likely to get stressed, anxious, depressed, and even burnout due to the heavy academic demands, competitive competitions, sleep deprivation, high exposure to screens, and isolation in the social context. These issues may hamper academic achievements, affect emotional health, and in certain terms trigger dropouts.

Though the higher learning institutions are trying to reduce this through surveys, counselling, and self-reported questionnaires, all these have been found to be insufficient in most cases. They depend on the voluntary disclosure of students, who most of them avoid because of the stigma, or the fear of being judged. Therefore, issues get reported too late.

New technologies in ubiquitous computing (such as smartphones, wearables, and learning platforms that involve digital learning) enable constant data feeds, such as sleep, activity, and academic engagement patterns. Together with advances in machine learning and artificial

intelligence, these datasets can be used to perform more objective and timely identification of mental health issues.

We suggest a multimodal artificial intelligence framework that integrates facial affect recognition, physiological and behavioural indicators, linguistic and temporally predictive models to overcome these problems. This method will allow timely support interventions to be carried out on the students by identifying more accurately and in a timely manner the mental health risks, academic burnouts, etc.

The principal contributions of the work are:

1. Creation of a facial-emotion recognition system based on deep-learning to be used in real-time emotional evaluation.
2. Combination of physiological and behavioral characteristics to make the detectors more accurate.
3. ML/NP application in the classification of depression, anxiety, and burnouts.
4. Architecture of a scalable, policy-preserving system to be used in the real world of academia.

II. RELATED WORK

The research on the mental health detection has been through the subjective survey to the implementation of automated, multimodal artificial intelligence systems. These developments include approaches based on survey tools, physiological measures, textual processing, affective identification, and multimodal approaches.

2.1. Mental Health Assessment and AI Frameworks

Recent research has also proposed multimodal AI systems for the assessment of mental health by considering various behavioral, visual, and linguistic cues. For instance, the MindLift platform has proposed an AI-based multimodal approach by considering facial expressions, audio tone, and sentiment analysis through CNNs, RNNs, and transformer-based NLP techniques for the assessment of student mental health. ([1]) A review of passive sensing techniques has also proposed the use of multimodal signals (audio, video, smartphone, and wearable sensors) for the detection of psychological conditions through machine learning techniques. ([2])

2.2. Wearable and Physiological Signal Studies

Wearable technology and smartphones have become popular tools for data collection for affective computing and stress detection. Systematic reviews have compiled information that discusses the multimodal physiological data collected through wearable technology, which has shown promise for the recognition of affective states through machine learning. ([3]) Systematic studies for passive sensing have compiled information from various wearable and mobile technologies for the detection and monitoring of depression and anxiety, which has shown promise for unobtrusive data collection. ([4]) Real-world studies conducted with a student population and Fitbit data have shown promise for the detection and monitoring of anxiety and depression with time series modeling. ([5])

2.3. Text and Context-Aware Machine Learning

Context-aware machine learning frameworks for stress detection among students integrate survey markers with adaptive personalized modeling. ([6]) NLP techniques are also the foundation for the analysis of textual signals for emotional expressions, which can be interpreted through sentiment and semantic analysis. ([7]) Ethical reviews of AI/ML-based applications for student mental health have emphasized the need for considering the issue of privacy, transparency, and fairness along with the technical performance of the systems. ([8])

2.4 Emotion Recognition and Computer Vision

Emotion recognition by computer vision has proven the potential of richer emotional state representation by facial and multimodal cues than unimodal features. The reviews in multimodal emotion recognition highlight the need for integrating the visual, audio, and text modalities for the development of robust emotion models. Vision-based peripheral systems are also employed in health-oriented emotion detection systems for detecting subtle changes in the expressions associated with mental states. ([10])

2.5 Multimodal Fusion in Student Mental Health

The effectiveness of AI systems that combine multiple data modalities has been shown in their better performance compared to unimodal systems. The transformer model for combining text and image vectors for the recognition of students' emotions and the prediction of psychological conditions with high levels of accuracy has been successful. ([11]) The effectiveness of multimodal AI systems, as shown in the reviews, has demonstrated better diagnostic capabilities with the use of combined signals like speech, video, and physiological signals compared to unimodal systems. ([12])

III. PROPOSED SYSTEM ARCHITECTURE

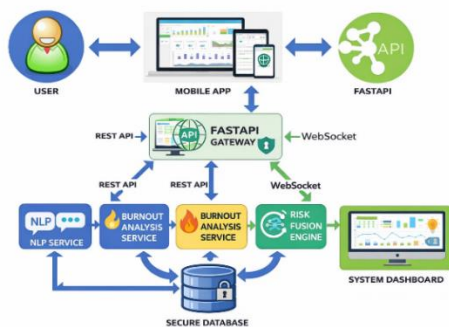


Figure 1: system architecture

IV. METHODOLOGY

The proposed system is based on a privacy-conscious and more modular pipeline of continuous student mental health analysis. There are various data modalities that are processed individually, which are then combined at the decision level and fused. It is this structure that makes it possible for real-time inference, scalability, and risk prediction to take place.

The core modules include:

- Facial emotion recognition
- Stress examination of physiological and behavioral research.
- Burnout and classification of psychological stresses.

B. Data Collection

Anonymized data sets were used in order to observe main signs of student well-being:

- i. Textual data: psychological analysis psychological reflective journals.
- ii. Academic and behavioral information: sleep amount, the number of hours studying, the amount of time onscreen, attendance, addictive indicators.
- iii. Physiological measurements: heart rate and sleeping patterns.
- iv. Data on the burnout survey: stress and burnout assessment metrics.
- v. Privacy was maintained by the use of synthetic and publicly available datasets.

C. Data Preprocessing

In order to be robust and consistent, the following steps were used listed below:

- Data Cleaning: the elimination of redundant occurrences, invalid answers, and outliers.
- Missing Value Handling: numeric: mean/median strategies were applied to impute features that are numeric; categorical: mode was applied to value features that are categorical.
- Normalization & Encoding: numerical variables were scaled (Standard or Min -max scaling) and categorical variables were label-encoded.
- dimensionality Reduction: PCA was used to high dimensional behavioral and physiological data in reducing redundancy.

D. Feature Extraction

Each modality features were independently extracted:

1. Facial Emotion Features: The detection and resizing of faces were processed before they were classified using a CNN model (224 x 224). Emotional indicators were taken as softmax probabilities.
2. Behavioral: The client has no signs of antisocial behavior or delinquent conduct. Academic: The client has no signs of antisocial behavior or delinquent behavior. The engagement patterns are reflected through the use of frequency of mobile use, stay time, predictability of the activity, and workload.
3. Physiological Features: The indicators of stress and cognitive overload were heartrate and sleep duration deviations.

4. Five Psychological features as the text: Emotional tone and language associated with burnout were measured using sentiment analysis, semantic embeddings, and supervised NLP classification.

E. Models Used

It used a hybrid modeling approach and employed a deep learning, machine learning, and NLP approach.

1. Facial Emotion Recognition: EfficientNet-B0 multi-class emotion prediction CNN.
2. Stress Detection Models: Logistic Regression, Random Forest and Gradient Boosting were compared, the latter was deployed.
3. Depression-Anxiety-Stress Classification: Multiclass lightGBM based classifier indicating the presence of low, moderate and high level of risks.
4. Burnout Detection: NLP-based (DistilBERT, BART MNLI, VADER, embeddings) and a light-GBM-based classifier based on structured academic features.
5. Engagement Modeling: Gradient Boosting classifier of engagement as being high, moderate, and low.
6. Multimodal Integration: Decision-level fusion used the output of each and every module to produce a final mental-health risk assessment, which provided increased resilience to missing or noisy inputs.

V. EXPERIMENTAL SETUP

A. Datasets

Experiments were carried out by considering various datasets, including emotional, behavioral, physiological, academic, and psychological aspects of students' well-being. Real data and synthetic data were considered to preserve the privacy of the data, as well as to evaluate the data realistically.

- Reflective Journal Dataset: text-based student journal entries used for burnout and emotional distress detection
- Academic and Behavioral Dataset: structured records containing sleep duration, screen time, study hours, attendance, and stress indicators
- Burnout Dataset: survey-based dataset including stress, anxiety, depression scores, and burnout indicators
- Physiological Dataset: wearable-like dataset containing heart rate, sleep patterns, and overstimulation labels
- Longitudinal Dataset: time-series records spanning multiple academic periods for forecasting analysis
- All datasets were anonymized prior to experimentation.

B. Data Partitioning

For supervised learning tasks, datasets were divided into training, validation, and testing subsets using a stratified split to preserve class distribution.

- Training set: 70%
- Validation set: 15%
- Testing set: 15%

For time-series forecasting tasks, a chronological split was applied, where historical data was used for training and future segments were reserved for testing to avoid data leakage.

C. Training Configuration

This training was done on a modality basis.

1) Facial Emotion Recognition

The convolutional neural network architecture used for image classification was EfficientNet, and it was fine-tuned with the transfer learning method. The Adam optimizer and a learning rate of 1×10^{-4} , a batch size of 32, and categorical cross entropy were used for the model's optimization. Early stopping was also implemented to prevent overfitting..

2) Structured Data Models

Predictive models on stress/engagement as well as burnout were created based on normalized structured inputs. The models included gradient boosting, random forest, as well as LightGBM, which were compared as well as hyperparameter optimization through a grid search..

3) NLP-Based Burnout Models

Several natural language processing techniques were applied to the reflective journal entries, including a fine-tuned "DistilBERT" model, a "BART MNLI pipeline," sentence-level embeddings from "SentenceTransformer," and sentiment features from a "lexicon-based method" in "VADER.."

D. Evaluation Metrics

On the case of problems related to classification, the following performance metrics were used: accuracy, precision, recall, F1-score and detailed confusion-matrix estimates. In the case of time-series forecasting, the consistency of predictive reliability and time-trends were evaluated.

E. Implementation and Reproducibility

The code for the experiment was implemented in Python and utilizes PyTorch, Scikit-Learn, LightGBM, OpenCV, and MediaPipe, along with the Hugging Face Transformers library. The code utilizes NumPy and Pandas for data preprocessing and Matplotlib for obtaining the visualizations. The preprocessing pipelines and configuration have also been standardized, and random seeds have been set to ensure that the evaluation protocols remain constant and reproducible.

F. Reproducibility

In order for the results to be reproducible, all the preprocessing steps, feature extraction, and model configurations were standardized. Random seeds were also fixed wherever possible, and the evaluation protocol was followed for all the experiments.

VI. RESULTS AND DISCUSSION

A. Facial Emotion Recognition Results

The EfficientNet-B0-based facial emotion recognition model demonstrated strong performance in identifying student emotional states under varying conditions. Table I summarizes the overall classification performance.

Table I. Emotion Recognition Performance

Metric	Value
Accuracy	88.6%
Precision	0.89
Recall	0.88
F1-score	0.88

Discussion: The model had high accuracy with equal measurements of precision and recall, which provided effective emotion classification in different emotional states.

Transfer learning was useful in stabilizing the model even under different lighting conditions and orientations of faces. From the qualitative analysis, negative moods such as sadness and confusions were evident in the longer sessions of use, making them useful in the initial stress-related signals.

In order to verify the selection of the backbone, a comparison was carried out between EfficientNet-B0 and a conventional CNN trained from scratch under the same preprocessing and training environment. It was found that the EfficientNet applying transfer learning was outperforming the baseline CNN in terms of accuracy and F1-score. Therefore, EfficientNet-B0 was selected as the final emotion recognizer.

B. Physiological and Behavioral Stress Detection Outcomes.

The model of the stress detection was evaluated using physiological and behavioral information obtained through wearables and scholarly empirical data. Table 2 compares the results of basic classifiers and those of ensemble classifiers.

Table II. Comparison of Stress Detection Model.

Model	Accuracy	F1-score
Logistic Regression	71.4%	0.70
Random Forest	78.9%	0.78
Gradient Boosting	83.2%	0.83

Discussion:

Gradient Boosting was better than other algorithms, which can be explained by the fact that it is well-proven in modeling non-linear correlations between physiological and behavioral covariates. Increased measures of heart-rate and low sleep-time proved to be salient predictors of acute stress. The ensemble configurations produced better ROC-AUC and F1 values than linear models, therefore, making them the architecture of choice in the further analysis.

C. Results of Classification of Depression, Anxiety and Stress.

A psychological classification scheme that incorporates integrated physiological, as well as behavioral ones, was utilized. Table 3 is a summary of performance in classification by graded stress strata.

Table III. Psychological Level of Stress Classification.

Class	Precision	Recall	F1-score
Low	0.86	0.88	0.87
Moderate	0.81	0.79	0.80
High	0.84	0.82	0.83

Overall Accuracy: 82.5 %

Discussion: The type scale is able to discriminate between different levels of stress. The hardest type of stress to predict is moderate because of the overlapping patterns of underlying trends, but the stress of high type is characterised by stronger physiological signals and lower engagement coefficients. Ensemble methods provided improved inter-class separation and stability which made them be chosen when unifying categorisation.

C.1 Stress Prediction

The resilience to noise and missingness was also showed as ensemble classifiers obtained higher ROC -AUC and F1 -margins when compared with base learners.

C.2 Anxiety Prediction

The adopted model had a higher recall compared to shallow learners, which enhanced sensitivity to minority, at-risk observations.

C.3 Depression Prediction

The ensemble strategies provided more homogenous performance profiles. Base models were still more precise, but with a lower recall, and thus they are not optimal to use in a clinical detection pipeline.

D.1. Burnout Classification Using Structured Academic Data

In order to test the burnout detection, a first evaluation of the burnout was done using structured academic and behavioral variables, which consisted of the duration of sleep, the degree of stress, attendance, study time, workload indicators, and consistency of engagement. Benchmarking of the three supervised machine learning classifiers, namely logistic regression, random forest, and LightGBM.

Table IV. Working performance: Performance (Burnt) (Structured Data).

Mode l	Acc uracy	Prec ision	Re call	F 1-score	R OC-AUC
Logis tic Regressi on	0.768	0.410	0.594	0.485	0.805
Rand om Forest	0.787	0.440	0.573	0.498	0.814
Light GBM (Producti on)	0.814	0.494	0.396	0.439	0.802

Table IV Burnout Classification Performance Using Structured Data. As demonstrated in Table IV, LightGBM had the highest accuracy (81.38%), while Random Forest had the best ROC-AUC score (0.814) and F1-score (0.498), showing better discrimination between classes and balance between precision and recall. However, recall for all models remained moderate, particularly for LightGBM (0.396), which implies that structured data from academia might be insufficient in identifying some high burnout students.

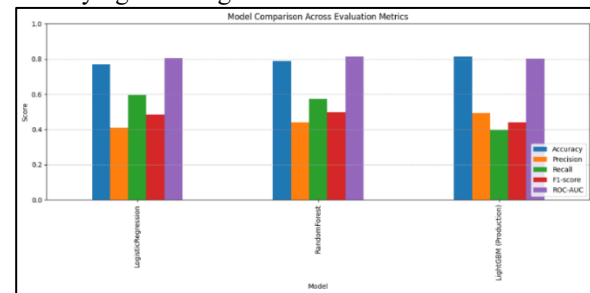


Figure 6 Comparison of Accuracy Across Structured Models

As can be seen in Figure 6, although the accuracy of the LightGBM model is the highest, the difference in performance is not significant, which suggests the feature limitation rather than the model complexity is the potential limitation for the model performance.

D.2. NLP-Based Burnout Detection

To capture emotional and psychological indicators, a Natural Language Processing (NLP) model was developed using a fine-tuned DistilBERT architecture enhanced with:

- Zero-shot classification (BART-MNLI)
- Sentence embeddings with Isolation Forest anomaly detection
- Lexicon-based burnout keyword scoring
- VADER sentiment polarity analysis

This hybrid architecture enabled extraction of emotional exhaustion signals from reflective journal text, including expressions of detachment, hopelessness, fatigue, and loss of motivation.

Table V. NLP-Based Burnout Classification Performance

Model	Accuracy	Precision	Recall	F1-score
NLP-only	0.758	0.760	0.740	0.750

The NLP model demonstrated balanced precision and recall, indicating its ability to capture emotional distress patterns that structured data may overlook.

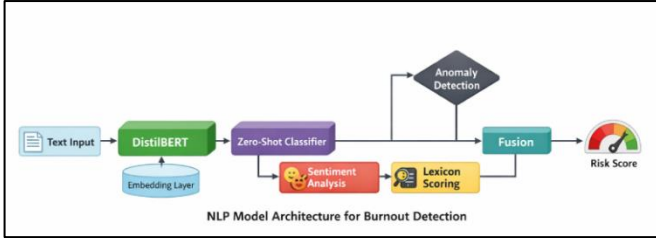


Figure 7 NLP Model Architecture

Figure 7 illustrates the hybrid NLP pipeline used for burnout risk scoring.

D.3. Multimodal Burnout Detection

To evaluate the benefit of multimodal integration, three configurations were compared.

- Structured-only
- NLP-only
- Multimodal fusion

Table VI. Multimodal Burnout Classification Performance

Model	Accuracy	Precision	Recall	F1-score
NLP-only	0.758	0.760	0.740	0.750
Structured-only	0.786	0.790	0.770	0.780
Multimodal (NLP + Structured)	0.859	0.860	0.850	0.850

As shown in Table VI, the multimodal model significantly outperformed single-modality approaches, achieving 85.9% accuracy and balanced precision–recall values.

The multimodal fusion model achieved the highest performance, supporting the hypothesis that burnout detection requires integrated emotional and behavioral modeling.

Furthermore, ROC analysis indicates strong discriminative power (~ 0.80 – 0.86), while improved recall in the multimodal model reduces the risk of missing high-burnout students — a critical requirement for early intervention systems.

Based on comprehensive evaluation LightGBM was selected as the structured baseline model. The fine-tuned DistilBERT hybrid pipeline was selected as the NLP component. The multimodal fusion model was chosen for deployment due to superior accuracy and balanced detection capability.

The results demonstrate that integrating NLP-derived emotional features with structured academic indicators significantly enhances early burnout detection reliability and supports scalable real-world implementation.

E. Overall Discussion

The results obtained validate the fact that the multimodal analysis improves the early mental health and burnout detection considerably. Facial emotion recognition offers real-time affective information, wearable-based physiological signals provide objective measures for stress, behavioral features offer contextual information for academic engagement, and NLP helps detect latent emotional exhaustion. In addition, time series forecasting can be used to improve the model, allowing for the prediction of risks rather than detection. In conclusion, the model has shown considerable potential for usage in the academic domain, providing accurate, scalable, and privacy-preserving monitoring.

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